



United International University

School of Science and Engineering

Mid Term Examination; Year 2025; Trimester: Summer

Course: PHY 2105/PHY 105; Title: Physics; Sec: All

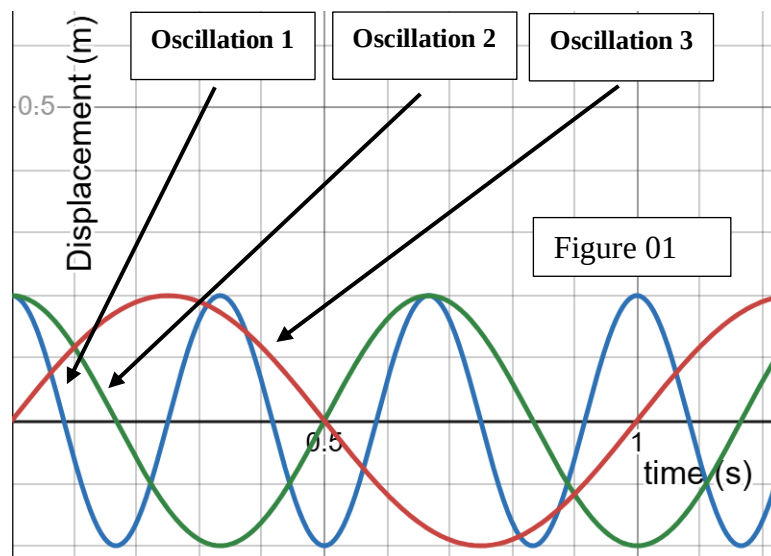
Full Marks: 30; Time: 1 Hour 30 Minutes

Questions no 1, 2 and 3 are mandatory to answer. Answer any one from question no 4 and 5.

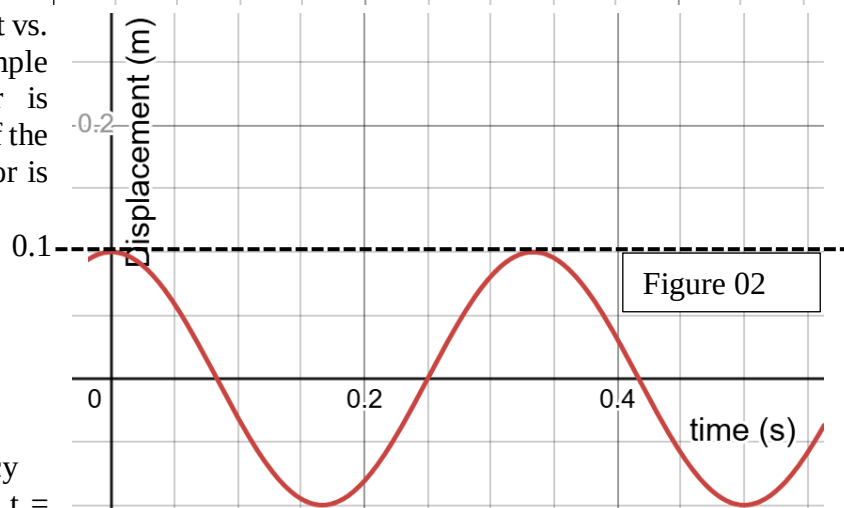
1. (a) Graphically show the phase difference between the two simple harmonic oscillators with the displacements $y_1 = A \cos \omega t$ and $y_2 = A \cos \left(\omega t + \frac{\pi}{4} \right)$ in the same frame of reference. 2 CO1

- (b) Why does the amplitude of the energy function of a Damped Harmonic oscillator decays faster than amplitude of displacement of oscillation? Use formula and logic as needed. 2 CO1

- (c) Three oscillations are shown in figure 01. Find out which particle of oscillation has maximum velocity? Justify your answer with proper logic. 2 CO1



2. (a) The displacement vs. time graph of a simple harmonic oscillator is given in figure 02. If the mass of the oscillator is 0.5 kg, 3 CO3



calculate

- (i) Angular frequency
(ii) Displacement at $t = 0.7$ s
(iii) Acceleration at $t = 0.06$ s.

- (b) The equation of displacement of a mass-spring system is, 3 CO3

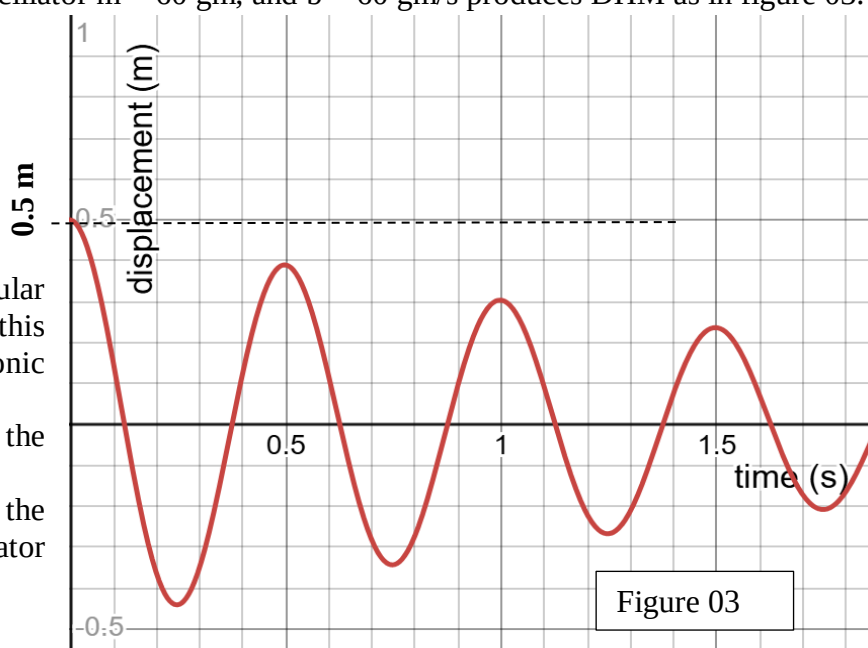
$$x = 20 \cos(155t + \frac{\pi}{6}) \text{ m}$$

The mass attached to the spring is 100 g. Calculate the (i) velocity at $t = 0.3$ s, (ii) total energy, and (iii) potential energy at $t = 0.5$ s.

3. (a) In an RLC circuit, $C = 0.2 \mu\text{F}$, $L = 20 \text{ mH}$, and $R = 180 \Omega$. 3 CO3
 (i) If oscillatory, find the frequency of oscillation.
 (ii) What is its life time, if any?
 (iii) For what resistance the motion will be critically damped?

- (b) For a damped oscillator $m = 60 \text{ gm}$, and $b = 60 \text{ gm/s}$ produces DHM as in figure 03. 3 CO3

- (i) What is the angular frequency of this Damped harmonic oscillator?
 (ii) What will be the amplitude at $t = 2 \text{ s}$?
 (iii) What will be the energy of the oscillator at $t = 2.5 \text{ s}$?



- (c) The displacement of a particle of a traveling wave is given by the equation, 4 CO3
 $y = 10\sin[\pi(36t - 0.001x)] \text{ m}.$

Calculate the (i) wave velocity, (ii) wavelength, (iii) frequency, (iv) particle speed at $t = 0.2 \text{ s}$, $x = 0.1 \text{ m}$.

4. (a) A mass spring system oscillates in SHM. The equation of displacement is, 4 CO2
 $y = A\sin(\omega t + \phi).$

- (i) Show that total energy, $E \propto A^2$.
 (ii) Graphically represent the potential energy and kinetic energy as a function of time where phase constant is 0.

- (b) A mass spring system is oscillating in DHM. 4 CO2

- (i) Derive the differential equation of DHM for the mechanical system.
 (ii) Write down the condition and possible solution of the differential equation for underdamped(oscillatory) damping.

5. (a) Show that for a traveling or progressive wave moving through a string, the differential equation of the wave motion is $\frac{d^2y}{dt^2} = v^2 \frac{d^2y}{dx^2}$ where the symbols have their conventional meanings. 4 CO2

- (b) Derive the differential equation of Simple Harmonic Motion for a simple pendulum and also show that 4 CO2

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Here the symbols have their usual meanings.

CO1: Define different physical quantities with examples. CO2: Derive/Show the various equations of SHM, DHM, wave motion, etc. CO3: Evaluate different numerical problems based on the basic characteristics of SHM, DHM, etc.